

V Abarajithan

Ramanathapuram

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[LinkedIn](#) [GitHub](#)

Professional Summary

Dynamic AI/ML Engineer currently at InfiniteTechAI, where I successfully transitioned from an intern to a full-time position by consistently delivering scalable AI solutions. I specialize in Self-RAG architectures, model fine-tuning, and end-to-end MLOps. With hands-on experience in building Docker images and establishing automated CI/CD pipelines for production-grade ML models, I am actively seeking a challenging new role to apply my technical expertise and drive AI innovation.

Certificates & Achievements

- **Advanced Data Science**, FITA Academy Madurai (April–July 2025) — ML, Stats, SQL, Python
- Member of Student Service Committee — Coordinated student outreach and support

Experience

Jr AI/ML Engineer(Intern – Full Time)
West Mambalam, Chennai

Infinite Tech AI, Chennai (Onsite)
Oct 9, 2025 – Present

Skills

- **Languages:** Python
- **Machine Learning:** Model Evaluation, SMOTE, Ensemble Methods, Collaborative Filtering
- **Natural Language Processing:** Tokenization, Lemmatization, Stopword Removal, Text Preprocessing
- **Data Analysis & Visualization:** EDA, Matplotlib, Seaborn, YData Profiling, NumPy, Pandas
- **Frameworks & Libraries:** Scikit-learn, LangChain , Pytorch, Llama-Index , Langgraph
- **AI Architectures:** RAG, LLaMA, Embedding Models, Vector Databases
- **Deployment Tools:** Flask, Streamlit, Git, GitHub, Jupyter Notebook, Docker , Jenkins , kubernetes

Projects

Self-RAG: AI Research Agent

github.com/Ajan2k/Self-RAG

FastAPI, LangGraph, LLMs, FAISS, WebSockets

- Engineered a self-reflective AI agent using LangGraph that autonomously critiques and revises its own responses to improve accuracy.
- Integrated a hybrid retrieval system combining local FAISS vector search with automatic web search escalation for missing context.
- Built a FastAPI WebSocket backend to stream the agent’s real-time reasoning and workflow steps directly to a custom frontend UI.

Customer Churn Prediction System

github.com/Ajan2k/customer-churn

CatBoost, Github Actions, Trivy, Docker

- Containerized the backend and a React/Vite frontend using secure, multi-stage **Docker** builds running as non-root users.
- Engineered a **GitHub Actions CI/CD pipeline** featuring automated **pytest** suites, **Ruff** linting, **Trivy** container scanning, and automated deployment to GHCR.

Fine-Tuning Pipeline: Qwen 2.5-7B LoRA

github.com/Ajan2k/fine-tuned-chatbot

LoRA, Unsloth, 4-bit optimization, SFT, mixed-precision

- Engineered an end-to-end LoRA fine-tuning pipeline for a 7B instruction model, enabling adapter-only training with 4-bit loading and mixed-precision optimizations to minimize GPU memory and accelerate iteration.
- Built reproducible training and evaluation workflows with centralized hyperparameter configuration, automated training/eval loss tracking, and inference routines for quantitative and qualitative validation.

Education

The American College, BCA

GPA: 8.1

Jul 2022 – May 2025

D.D. Vinayagar Hr Sec School, Bio Maths

Percentage: 75.5%

Jun 2020 – Mar 2022

The Government School, Sholandur

Percentage: 81.5%

Jun 2015 – Apr 2020